Întrebarea 536

Rezolvare

$$n = 113$$

Verificăm relația lui Fermat pentru cel mult 3 martori.

$$b \in \{0, 1, ..., 112\}$$

Alegem martorul $b = 2$.

$$2^{112} = (2^2)^{56} = 4^{56} = (4^2)^{28} = 16^{28} = (16^2)^{14} = 256^{14} \equiv 30^{14} \pmod{113}$$
$$30^{14} = (30^2)^7 = (900)^7 \equiv (109)^7 \equiv (-4)^7 \pmod{113}$$
$$(-4)^7 = -4 \cdot (-4)^6 = -4 \cdot (16)^3 = -4 \cdot 16 \cdot (16)^2 = -4 \cdot 16 \cdot 256 \equiv -4 \cdot 16 \cdot 30 \pmod{113}$$
$$16^3 = 16 \cdot 30 = 480 \equiv 28 \pmod{113}$$
$$-4 \cdot 16 \cdot 30 = -4 \cdot 480 \equiv -4 \cdot 28 \pmod{113}$$
$$(-4) \cdot 28 = -112 \equiv 1 \pmod{113}$$

$$2^{112} \equiv 1 \pmod{113}$$

Alegem martorul $b = 3$.

$$3^{112} = (3^2)^{56} = 9^{56} = (9^2)^{28} = 81^{28} = (81^2)^{14} = 6561^{14} \equiv 7^{14} \pmod{113}$$
$$7^{14} = (7^2)^7 = 49^7 = 49 \cdot 49^6 = 49 \cdot 2401^3 \equiv 49 \cdot 28^3 \pmod{113}$$
$$49 \cdot 28^3 = 49 \cdot 28 \cdot 28^2 = 49 \cdot 28 \cdot 784 \equiv 49 \cdot 28 \cdot 106 \equiv 49 \cdot 28 \cdot (-7) \pmod{113}$$
$$49 \cdot 28 \cdot (-7) = 1372 \cdot (-7) \equiv 16 \cdot (-7) \pmod{113}$$
$$16 \cdot (-7) = -112 \equiv 1 \pmod{113}$$

$$3^{112} \equiv 1 \pmod{113}$$

Alegem martorul $b = 5$.

$$5^{112} = (5^2)^{56} = 25^{56} = (25^2)^{28} = 625^{28} \equiv 60^{28} \pmod{113}$$
$$60^{28} = (60^2)^{14} = 3600^{14} \equiv 97^{14} \pmod{113}$$
$$97^{14} = (97^2)^7 = 9409^7 \equiv 30^7 \pmod{113}$$
$$30^7 = 30 \cdot 30^6 = 30 \cdot (30^2)^3 = 30 \cdot 900^3 \equiv 30 \cdot 109^3 \equiv 30 \cdot (-4)^3 \pmod{113}$$
$$30 \cdot (-4)^3 = 30 \cdot (-4) \cdot (-4)^2 = 30 \cdot (-4) \cdot 16 = -120 \cdot 16 \equiv 106 \cdot 16 \pmod{113}$$
$$106 \cdot 16 = 1696 \equiv 1 \pmod{113}$$

$$5^{112} \equiv 1 \pmod{113}$$

$$2^{112} \equiv 1 \pmod{113}, \quad 3^{112} \equiv 1 \pmod{113}, \quad 5^{112} \equiv 1 \pmod{113}$$

Rezultă că 113 trece testul lui Fermat pentru cei trei martori aleși.